

Statement 1 (Coin tosses). *A fair coin is tossed twice. Let X be the number of heads and let $Y = 1$ if the outcomes are equal and $Y = 0$ otherwise. Find $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, and $\text{Cov}(X, Y)$. Are X and Y independent?*

Statement 2 (Discrete joint distribution). *The joint distribution of X and Y is*

$Y \backslash X$	-1	0	1
0	$\frac{1}{12}$	$\frac{2}{12}$	0
1	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{1}{12}$
2	0	$\frac{2}{12}$	$\frac{3}{12}$

Find the marginal distributions, expectations, and variances of X and Y . Also find $\Pr(X + Y \geq 2)$.

Statement 3 (Pareto distribution). *Let*

$$f_X(x) = \begin{cases} c/x^4, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

Find c , $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var}(X)$.

Statement 4 (Poisson distribution). *Let*

$$\Pr(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Find $\mathbb{E}[X]$, $\mathbb{E}[X(X - 1)]$, and $\text{Var}(X)$.

Statement 5 (Random Gaussian matrix). *Let $W \in \mathbb{R}^{m \times n}$ have independent entries $W_{ij} \sim \mathcal{N}(0, \sigma^2)$. Find*

$$\mathbb{E}[\|W\|_F^2], \quad \text{Std}(\|W\|_F^2), \quad \frac{\text{Std}(\|W\|_F^2)}{\mathbb{E}[\|W\|_F^2]}.$$

What happens to the relative standard deviation as mn increases?

Statement 6 (Attention scaling). *Let $q, k \sim \mathcal{N}(0, I_d)$ be independent and define*

$$z = \frac{\langle q, k \rangle}{c}.$$

Find $c > 0$ such that $\text{Var}(z) = 1$.

Bonus Tasks

Statement 7 (Stochastic-gradient noise). Let $x_i \sim \mathcal{N}(0, 1)$ and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ be mutually independent. Define

$$L_B(\theta) = \frac{1}{2B} \sum_{i=1}^B ((\theta - \theta_*)x_i - \varepsilon_i)^2.$$

Find

$$\text{Std}\left(\frac{\partial L_B(\theta)}{\partial \theta}\right).$$

You may use $\mathbb{E}[x_i^2] = 1$ and $\mathbb{E}[x_i^4] = 3$.

Statement 8 (Kaiming initialization). Let $x_i \sim \mathcal{N}(0, 1)$ and $w_i \sim \mathcal{N}(0, \sigma_w^2)$ be mutually independent. Define

$$z = \sum_{i=1}^n w_i x_i, \quad h = \max(0, z).$$

Find $\mathbb{E}[z]$, $\mathbb{E}[z^2]$, and $\mathbb{E}[h^2]$. Find σ_w^2 such that $\mathbb{E}[h^2] = 1$.

Statement 9 (Standard normal distribution). Let $X \sim \mathcal{N}(0, 1)$. Compute

$$M(t) = \mathbb{E}[e^{tX}],$$

and find $M^{(n)}(0)$ for every integer $n \geq 0$.

Statement 10 (BatchNorm variance). Let $X_1, \dots, X_B$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$. Define

$$\bar{X} = \frac{1}{B} \sum_{i=1}^B X_i, \quad S_B^2 = \frac{1}{B} \sum_{i=1}^B (X_i - \bar{X})^2.$$

Find $\mathbb{E}[\bar{X}]$, $\text{Var}(\bar{X})$, and $\mathbb{E}[S_B^2]$. Modify S_B^2 to obtain an unbiased estimator of σ^2 .

Answers

1. $\mathbb{E}[X] = 1$, $\mathbb{E}[Y] = \frac{1}{2}$, $\mathbb{E}[XY] = \frac{1}{2}$, and $\text{Cov}(X, Y) = 0$. They are not independent.
- 2.

$$\Pr(X = -1, 0, 1) = \left(\frac{2}{12}, \frac{6}{12}, \frac{4}{12}\right), \quad \mathbb{E}[X] = \frac{1}{6}, \quad \text{Var}(X) = \frac{17}{36}.$$

$$\Pr(Y = 0, 1, 2) = \left(\frac{3}{12}, \frac{4}{12}, \frac{5}{12} \right), \quad \mathbb{E}[Y] = \frac{7}{6}, \quad \text{Var}(Y) = \frac{23}{36}.$$

Moreover, $\Pr(X + Y \geq 2) = \frac{5}{12}$.

3.

$$c = 3, \quad \mathbb{E}[X] = \frac{3}{2}, \quad \mathbb{E}[X^2] = 3, \quad \text{Var}(X) = \frac{3}{4}.$$

4.

$$\mathbb{E}[X] = \lambda, \quad \mathbb{E}[X(X - 1)] = \lambda^2, \quad \text{Var}(X) = \lambda.$$

5.

$$\mathbb{E}[\|W\|_F^2] = mn\sigma^2, \quad \text{Std}(\|W\|_F^2) = \sigma^2\sqrt{2mn}, \quad \frac{\text{Std}}{\mathbb{E}} = \sqrt{\frac{2}{mn}}.$$

The relative standard deviation converges to zero as $mn \rightarrow \infty$.

6.

$$c = \sqrt{d}.$$

7.

$$\text{Std}\left(\frac{\partial L_B(\theta)}{\partial \theta}\right) = \sqrt{\frac{2(\theta - \theta_*)^2 + \sigma^2}{B}}.$$

8.

$$\mathbb{E}[z] = 0, \quad \mathbb{E}[z^2] = n\sigma_w^2, \quad \mathbb{E}[h^2] = \frac{n\sigma_w^2}{2}, \quad \sigma_w^2 = \frac{2}{n}.$$

9.

$$M(t) = e^{t^2/2}, \quad M^{(n)}(0) = \begin{cases} 0, & n \text{ odd}, \\ \frac{(2k)!}{2^k k!} = (2k-1)!!, & n = 2k. \end{cases}$$

10.

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{B}, \quad \mathbb{E}[S_B^2] = \frac{B-1}{B}\sigma^2.$$

An unbiased estimator is

$$\frac{1}{B-1} \sum_{i=1}^B (X_i - \bar{X})^2.$$